DEPARTMENT OF Electrical Engineering
UNIVERSITY OF ENGINEERING & TECHNOLOGY, PESHAWAR
JALOZAI CAMPUS

Course: Computer Programming (2nd Semester)
Assignment: Assignment No. 1 (Spring-2026)
Student Name: MUHAMMAD MUSA
Registration No :BF25NWELE0754

----

i. Variable Declaration :

```
#include <iostream> using
namespace std;

int main() {
    // Defining these as double since electrical values often have decimals
double voltage, current, resistance;     return 0;
}
```

ii. Display Message :

```
#include <iostream> using
namespace std;

int main() {
    // Just a simple print statement to welcome the user
cout << "Welcome to Object Oriented Programming in C++!";
return 0;
```

} **iii. Read**

Age :

```
#include <iostream> using
namespace std;
 int main() {     int age;     cout << "Please enter your age:
";     cin >> age; // Saving the user input into the age
variable     return 0;
}
```

iv. Temperature Check :

```
#include <iostream> using
namespace std;
 int main()
{     int
temperature
= 105; //
Testing with
```

```
a value over
100     if
(temperature
> 100) {
cout <<
"Overheating
detected!";
// This only
runs if it's
too hot
    }
return 0;
}
```

v. Area of a Circle :

```
#include <iostream> using
namespace std;
 int main() {     double
r = 5.0, area;
    // Using the constant PI value given in the assignment
area = 3.14159 * r * r;     cout << "The area of the
circle is: " << area;     return 0;
}
```

vi. Print x, y, and

z :

```
#include <iostream> using
namespace std;
 int main() {     int x = 10, y
= 20, z = 30;
    // Printing them all out with commas in between
cout << x << ", " << y << ", " << z;     return 0;
}
```

vii. Sum of Voltage and Current :

```
#include <iostream> using
namespace std;
 int main() {     double voltage = 12.0, current =
0.5, total;     total = voltage + current; // Adding
them together     return 0;
}
```

viii. Division by Zero Guard :

```
#include <iostream> using
namespace std;
 int main()
{
    int denominator = 0;
    // Safety check to make sure we don't divide by zero
if (denominator == 0) {         cout << "Error: Division
by zero";
    }
return 0;
```

```
}
```

ix. Increment and Print :

```
#include <iostream> using
namespace std;
 int main() {
int count = 10;
    // Incrementing first then printing the new value
cout << ++count;
    return 0;
}
```

x. Read Three Floats :

```
#include <iostream> using
namespace std;
 int main() {     float a, b, c;     cout << "Enter
three floating-point numbers: ";     cin >> a >> b
>> c; // Getting all three at once     return 0;
```

} **xi. Total Resistance**

(Series) :

```
#include <iostream> using
namespace std;
 int main() {     double r1, r2, r3, total;     cout
<< "Enter values for 3 series resistors: ";     cin
>> r1 >> r2 >> r3;     total = r1 + r2 + r3; //
Series is just the sum     cout << "Total Series
Resistance: " << total;     return 0;
```

} **xii. Total Resistance**

(Parallel) :

```
#include <iostream> using
namespace std;
 int main() {     double r1, r2, r3, total;     cout
<< "Enter values for 3 parallel resistors: ";     cin
>> r1 >> r2 >> r3;
    // Using the reciprocal formula for parallel circuits
total = 1.0 / (1.0/r1 + 1.0/r2 + 1.0/r3);     cout <<
"Total Parallel Resistance: " << total;     return 0;
}
```

xiii. Voltage Division Rule (VDR) :

```
#include <iostream> using
namespace std; int main()
{
    // Picking some standard values for the circuit
double R1 = 50, R2 = 100, V_source = 15, VR2;
    // VDR formula: (Source Voltage * R2) / Total Resistance
VR2 = (V_source * R2) / (R1 + R2);     cout << "The voltage
across R2 is: " << VR2;     return 0;
}
```

xiv. Current Division Rule (CDR) :

```
#include <iostream> using
namespace std;
 int main()
{
    // Selecting resistors and total current
double R1 = 10, R2 = 40, I_total = 2.5, IR2;
    // CDR formula: (Total Current * R1) / (R1 + R2)
IR2 = (I_total * R1) / (R1 + R2);     cout << "The
current through R2 is: " << IR2;     return 0;
}
```

xv. Semester GPA Calculation :

```
#include <iostream> using
namespace std;
 int main() {     double g1, g2, g3, c1, c2, c3, gpa;     cout << "Enter
GPA and credits for 3 subjects: ";     cin >> g1 >> c1 >> g2 >> c2 >> g3
>> c3;     // Multiplying each grade by its credits and dividing by total
credits     gpa = (g1*c1 + g2*c2 + g3*c3) / (c1 + c2 + c3);     cout <<
"My semester GPA is: " << gpa;     return 0;
}
```